

Q1)

Known

$$T_1 = 300^\circ\text{K}$$

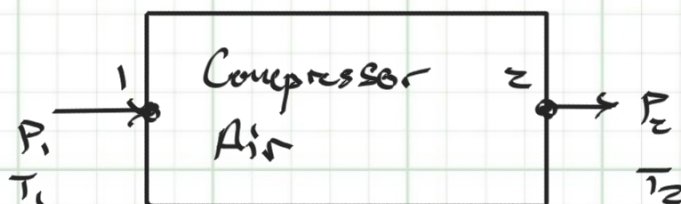
$$P_1 = 100 \text{ kPa}$$

$$T_2 = 460^\circ\text{K}$$

$$P_2 = 5 \text{ MPa}$$

$$\eta_c = \frac{1W_{2s}}{1W_2} = \frac{h_1 - h_{2s}}{h_1 - h_2}$$

Schematic



Assumptions

S.S, open

Adiabatic

Neglect $\Delta PE, \Delta KE$

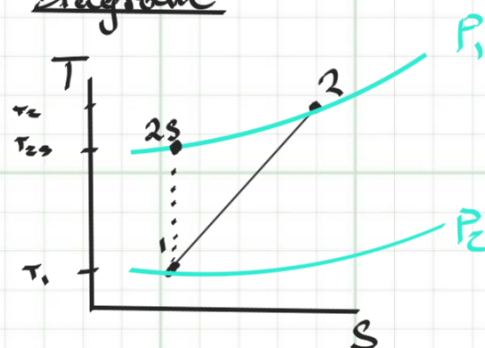
I.G.

Find

$$\eta_c$$

$$\sigma_m$$

Diagram



Symbolic

C.O.M: $\dot{m}_1 = \dot{m}_2 = \dot{m}$

1st law: $0 = \dot{Q} - \dot{W} + \dot{m}(h_1 - h_2)$

$$\Rightarrow \dot{W}_{in} = \dot{W}_2 = \dot{m}(h_1 - h_2)$$

2nd law: $0 = \frac{\dot{Q}}{T_B} + \dot{m}(s_1 - s_2) + \dot{\sigma}$

$$\Rightarrow \sigma_m = s_2 - s_1 \quad \text{For reversible isentropic process } s_2 = s_1$$

$$s_2 - s_1 = s_2^\circ - s_1^\circ - R \ln\left(\frac{P_2}{P_1}\right) \Rightarrow \sigma_m = s_2^\circ - s_1^\circ - R \ln\left(\frac{P_2}{P_1}\right)$$

Numeric

From Tables

$$h_1 = 300.19$$

$$h_2 = 1600.55$$

$$s_1^\circ = 1.70203$$

$$s_2^\circ = 2.42128$$

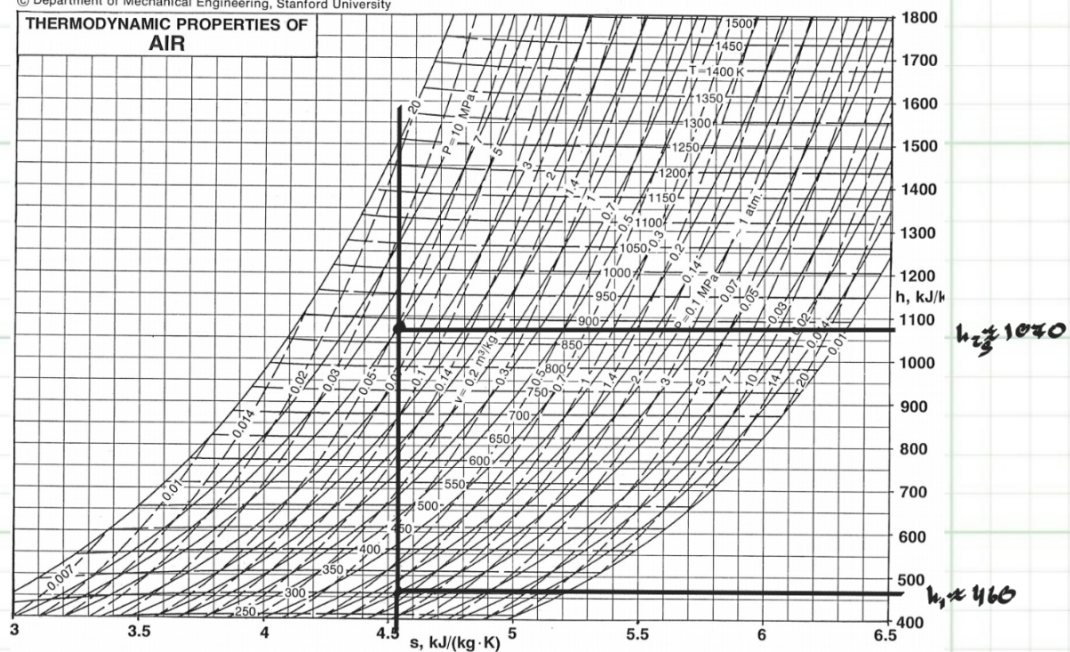
$$R = \frac{\bar{R}}{M_{Air}} = \frac{8.3143 \text{ kJ/kmol}\cdot\text{K}}{28.97 \text{ kg/kmol}} = 0.286997 \text{ kJ/kg}$$

$$\sigma_m = (2.42128 - 1.70203) - 0.286997 \ln\left(\frac{5 \text{ MPa}}{0.1 \text{ MPa}}\right)$$

$$\sigma_m = 0.09651 \text{ kJ/kg Ideal Gas}$$

$$1W_2 = h_1 - h_2 = 300.19 - 1600.55 = -1300.36$$

* h_{2s} can not be found from tables



$$w_{zs} = h_1 - h_{zs} = 460 - 1040 = -610$$

$$\therefore \eta_c \approx \frac{-610}{-700.36} \approx 87.1\%$$

Q2)

Known

$$T_1 = 650^\circ \text{K}$$

$$P_1 = 5 \text{ MPa}$$

$$P_2 = 2 \text{ MPa}$$

$$\eta_T = 90\% = \frac{w_{\text{net}}}{w_{\text{net},s}} = \frac{h_1 - h_2}{h_1 - h_{2s}}$$

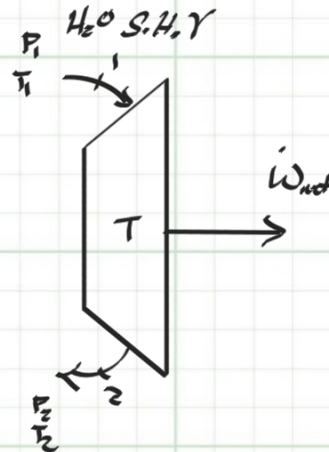
Assumptions

S.S., open sys.

Neglect ΔPE , ΔKE

Adiabatic

Schematic/Diagram



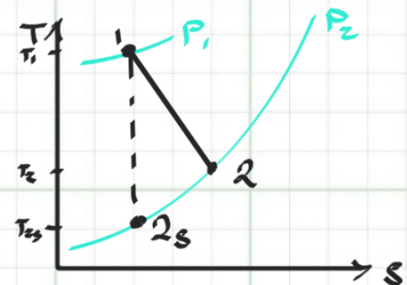
Determine

$$h_{2s}$$

$$T_{2s}$$

$$h_2$$

$$T_2$$



Symbolic

$$\text{C.O.M.: } m_1 - m_2 = 0 \Rightarrow \dot{m}_1 = \dot{m}_2 = \dot{m}$$

$$\text{C.O.E.: } 0 = \dot{Q} - \dot{W} + \dot{m}(h_1 - h_2)$$

$$\Rightarrow \dot{W}_{\dot{m}} = w_{\text{net}} = h_1 - h_2 \text{ for Real Process}$$

$$\Rightarrow w_{\text{net},s} = h_1 - h_{2s} \text{ for Isentropic Process}$$

$$\Rightarrow \eta_T = 90\% = \frac{w_{\text{net}}}{w_{\text{net},s}} = \frac{h_1 - h_2}{h_1 - h_{2s}} \Rightarrow h_2 = h_1 + 0.9(h_{2s} - h_1)$$

For Change in state I.C.,

$$\left. \begin{aligned} h_1 - h_2 &= c_{p,h_2}(T_1 - T_2) \\ h_1 - h_{2s} &= c_{p,h_2}(T_1 - T_{2s}) \end{aligned} \right\} \Rightarrow \eta_T = \frac{T_1 - T_2}{T_1 - T_{2s}} \Rightarrow T_2 = T_1 + 0.9(T_{2s} - T_1)$$

h_{2s} & T_{2s} are found using Tables @ given/known state Properties

Numeric

From Table

$$@ T = 350^\circ \text{C} @ 5 \text{ MPa} \quad h = 3068.4 \text{ kJ/kg} \quad s = 6.4493 \text{ kJ/kg K}$$

$$@ T = 400^\circ \text{C} @ 5 \text{ MPa} \quad h = 3195.6 \text{ kJ/kg} \quad s = 6.6459 \text{ kJ/kg K}$$

From Interpolation $| T = 650^\circ K = 376.85^\circ C$

$$h_1 = 3068.4 + (376.85 - 350) \frac{(3195.7 - 3068.4)}{(400 - 350)}$$

$$s_1 = 6.4493 + (376.85 - 350) \frac{(6.6459 - 6.4493)}{(400 - 350)}$$

$$h_1 = 3136.76 \text{ kJ/kg}$$

$$s_1 = 6.5548 \text{ kJ/kg K}$$

for Isentropic Process $s_1 = s_2$

$$\therefore T_{2s} = T @ s_1 \mid P = 2 \text{ MPa S.H.V}$$

$$h_{2s} = h @ s_1$$

From Table @ 2 MPa

$$@ T = 240^\circ C \quad s = 6.4952 \quad h = 2876.5$$

$$@ T = 280^\circ C \quad s = 6.6828 \quad h = 2976.4$$

From Interpolation $| s = 6.5548 \text{ kJ/kg K}$

$$T_{2s} = 240 + \frac{(6.5548 - 6.4952)(280 - 240)}{(6.6828 - 6.4952)}$$

$$h_{2s} = 2876.5 + \frac{(T_{2s} - 240)(2976.4 - 2876.5)}{280 - 240}$$

$$T_{2s} = 252.4^\circ C \approx 525.4^\circ K$$

$$h_{2s} = 2908.2 \text{ kJ/kg}$$

$$h_2 = 3136.76 + 0.9(2908.2 - 3136.76)$$

$$h_2 = 2931.1 \text{ kJ/kg}$$

$$T_2 = 680 + 0.9(525.4 - 680)$$

$$T_2 = 538.3^\circ K$$

